



Advances in renewable and sustainable energy technologies

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The transition to renewable and sustainable energy sources is a global imperative in the fight against environmental degradation and climate change. The depleting fossil-fuel reserves and rising energy demand pose challenges to energy security and contribute to increasing concerns over energy costs. Additionally, emissions from conventional energy resources contribute significantly to environmental deterioration and raise grave health concerns. Greenhouse gases generated by these energy resources aggravate global warming, while NO_x and particulate matter are linked to respiratory ailments. To address these pressing issues, research efforts are being directed toward the development of cleaner energy technologies, renewable feedstocks, advanced catalysts and electrocatalysts, sustainable energy-storage systems, waste valorization strategies, and environmentally responsible production processes.

The second edition of International Conference on Novel Materials and Technologies for Energy and Environment Applications (NMTE2A-2024) was organized by Birla Institute of Technology and Science, Pilani Hyderabad Campus from 17 to 18 February 2024. The conference provided an interdisciplinary forum encompassing diverse aspects of emerging materials, energy technologies, and environmental applications. It was our privilege to welcome more than 250 participants during the conference. The conference featured leading scientists from academia and industry who discussed recent developments, emerging challenges, and future directions in energy and environmental research. More than 200 submissions were received for NMTE2A-2024, alongside 5 keynote speeches and 20 invited talks. Following the conference, selected contributions were invited for consideration in this Special Issue, subject to the journal's peer-review and editorial processes. Forty-five articles were submitted to this issue, of which nine were accepted for publication following rigorous peer review.

The Special Issue entitled “Advances in Renewable and Sustainable Energy Technologies” brings together nine research articles addressing interconnected aspects of sustainable energy, environmental protection, resource recovery, and sustainable materials. Collectively, the contributions reflect the interdisciplinary nature of research at the intersection of energy, environmental sustainability, resource efficiency, and sustainable materials. The articles cover five broad and interconnected themes:

- Hydrogen and Catalytic Energy Conversion: Methane cracking for hydrogen production and catalytic conversion of biogas through dry reforming.
- Electrocatalysis and Sustainable Energy Materials: Development of photo/electrocatalytic materials for hydrogen evolution, oxygen reduction, and urea oxidation.
- Life-Cycle Assessment and Environmental Sustainability: Life-cycle assessment of energy-storage technologies and food-production systems to evaluate environmental impacts and sustainability.

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- **Waste Valorization and Resource Recovery:** Valorization of agricultural and industrial waste streams for the recovery of valuable products and resources.
- **Environmental Remediation and Sustainable Construction Materials:** Functional materials for multi-metal wastewater treatment and utilization of industrial residues in sustainable construction materials.

Together, these studies highlight advances in energy conversion, materials development, environmental assessment, and resource recovery, emphasizing the importance of coupling technological innovation with environmental and resource considerations to support the transition toward low-carbon, resource-efficient, and sustainable systems.

The Guest Editors sincerely thank all authors for their valuable contributions and the reviewers for their constructive comments and dedicated efforts in ensuring the quality of the published articles. We also gratefully acknowledge the editorial team of Environmental Science and Pollution Research for their support and guidance throughout the preparation and publication of this Special Issue.

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Prof. Iyman Abrar is an Assistant Professor of Chemical Engineering at BITS Pilani, Hyderabad Campus, India. She received her Ph.D. in Chemical Engineering from IIT Delhi in 2020. She has more than five years of experience in academia and research and has published over 30 articles in reputed international journals and conference proceedings. She is also an inventor on four Indian patents and is involved in sponsored research projects worth more than INR 1 crore from various

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Prof. Rajamohan Natarajan is a Professor of Chemical Engineering at Sohar University, Oman, with more than 25 years of teaching and research experience at various international higher education institutions. He has been recognized among the world's top 2% of scientists in the Stanford database. He has held several leadership roles, including Acting Dean and Assistant Dean of the Faculty of Engineering at Sohar University, and currently leads the University Research Theme on

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Prof. Mohammad Rehan is an Associate Professor at the Center of Excellence in Environmental Studies (CEES), King Abdulaziz University, Saudi Arabia, and has been recognized among the world's top 2% of scientists in the Stanford database. He has more than 20 years of experience in research and development, teaching, training, and project management. He pioneered waste-to-energy and sustainable energy systems research in Saudi Arabia and

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Prof. Luca Fiori is a Professor of Chemical Engineering at the University of Trento (UniTN), Italy, where he leads the Green Processes Engineering group and the Biomass Lab. He has been recognized among the world's top 2% of scientists in the Stanford database. He has more than 20 years of experience in academic teaching, research and development. Prior to joining UniTN, he worked for four years in two companies, one in the food-processing sector and the other in the design

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Prof. Inkollu Sreedhar is a Professor of Chemical Engineering at BITS Pilani, Hyderabad Campus, India, and has been recognized among the world's top 2% of scientists in the Stanford database. He received his Ph.D. in Chemical Engineering from BITS Pilani in 2008 and has more than 25 years of experience in industry, academia, and research. He has published over 108 articles in reputed international journals, four book chapters, one patent, and 60 conference proceedings. He has secured sponsored research projects worth more than INR 2.5 crore from various government agencies and industrial organizations. He has held several academic leadership positions at BITS Pilani and currently serves as Associate Dean, International Programmes and Collaborations Division, BITS Pilani Hyderabad Campus. He is a Fellow of the Telangana Academy of Sciences (FTAS), Fellow of the Royal Society of Chemistry (FRSC), Fellow of the Institution of Engineers (India) (FIE), and Senior Individual Member of the Indian National Academy of Engineering (INAE). His research interests include CO₂ capture and utilization, reaction engineering, heterogeneous catalysis, fluid rheology, and wastewater treatment.

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